Nishal Thomas

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Education

St. Mary's Sr. Sec. School — **New Delhi, India**

Class 11 — *Ongoing*

Class 10 — 91% — *2025*

Research and Technical Projects

Continuum [K2-Think Hackathon]

Hybrid AI Safety Infrastructure & Drift Monitor

- Built a Chrome extension to detect contradictions in LLM conversations using cumulative drift tracking and structural analysis (FastAPI backend)
- Implemented a cost-aware hybrid pipeline ($<$200ms local latency) that selectively invokes K2 Think V2 for formal contradiction adjudication

Mapping Hallucinations in Tiny Aya

Multilingual Hallucination Detection Study

- Participated in Expedition Tiny Aya, a global research initiative by Cohere Labs, collaborating within a 5-member international team to study multilingual hallucinations in Tiny Aya
- Applying black-box metrics (Hallucination Rate, Self Consistency, Variance, Cross-Model Disagreement, Prompt Sensitivity Score etc.) alongside white-box methods to detect hallucination signals using Python based pipelines

Step-Free Arithmetic Transformers

Mechanistic Interpretability Study

- Trained compact transformers (0.66M–10M params) using PyTorch, AdamW (FP16) on consumer hardware (RTX 3050 Laptop GPU)
- Performed attention head ablation and linear probing to identify structure-sensitive circuits and their causal role in generalization
- Built a reproducible training and analysis pipeline and authored a technical blog: Link

Grokking Feasibility Boundaries in Transformers

Replication and Analysis: Generalization Dynamics in Modular Arithmetic Transformers

- Trained transformers (3.2M params) on modular arithmetic using PyTorch, AdamW (FP16) on consumer hardware (RTX 3050) with controlled train–test overlap
- Analyzed representation rank and attention entropy to identify phase transitions during delayed generalization (grokking)
- Built an automated experimental pipeline with reproducible runs and published a blog: Link

Smart Traffic Light System [High School Science Fair]

Computer Vision + Embedded Controls

- Built a YOLOv11-based traffic control system (Python, OpenCV) with physical demonstration
- Integrated Arduino-based signal control with stabilization logic

Extracurriculars

- Participated in multiple speaking competitions including speech and debate, securing awards and recognitions at inter-school level.
- Served as a member of the School Student Council, contributing to student representation and initiatives
- Qualified Stage 1 of the Indian National Olympiad of Artificial Intelligence (INAIO)

Awards

2025: **YMCA Academic Excellence Award** — Class 10 Board performance

2025: **Science Topper — Class 10**, St. Mary's Sr. Sec. School

2025: **100/100 in Artificial Intelligence** — Board Examination